Supplementary Material for
Local Evidence Is Not Completion:
Evidence-Condition Geometry for Bounded Completion-Audit Workflows

Anonymous Submission

Scope of the Supplement

This anonymous supplementary material contains reproducibility and audit details that are intentionally kept out of the main technical-content pages. These tables support the evidence-condition controller evaluation but are not used as additional main effectiveness evidence. In particular, the workflow pilot is a workflow-shape/interface validation only, and the scalar singleton/Chao proxy is a negative control because it does not encode source-route evidence conditions.

Reproducibility Notes

Script entry points. All paths are relative to the repository root. The supplement and paper figures are assembled from `2/analysis/research_object_geometry/real_agent_pilot/` by running the Python entry point `credibility_supplement/run_credibility_supplement.py`. The frozen task logs used by this assembler are produced by `scripts/run_blind_policy_task.py`, `scripts/run_blind_code_task.py`, `external_validation_requests/run_external_requests_validation.py`, `controller_validation_v1/run_controller_validation_v1.py`, `controller_validation_v1/run_controller_validation_v2.py`, and `external_validation_v2/run_external_validation_v2.py`. The controlled stress tests are in `2/analysis/research_object_geometry/`: `controlled_geometry_simulation_v2.py` and `controlled_exposure_localization_v3.py`.

Seeds and thresholds. Generated-task method validation, external repository repair validation, and external boundary validation use 200 fixed seeds per challenger or controller policy before oracle scoring. Safe-state validation uses 6000 order/budget perturbations. The controller-validation thresholds are $\tau_s = 0.75$ for source-route support and $\tau_g = 0.70$ for exposure Gini. The 0.90 bounded-oracle recall threshold is used only for post-hoc false-certification evaluation and is not visible to the controller.

Data and log format. Primary run logs are JSONL action/evidence ledgers. Each event records a task identifier, source family, source-route stratum, route, action or scan record, discovered item identifier when present, new-item flag, and cost fields used for post-stop repair or confirmation-audit accounting. Aggregated CSVs in `credibility_supplement/results/` contain condition summaries, controller decisions, repair confidence intervals, seeded safe-state summaries, threshold/budget sweeps, oracle summaries, route-pattern examples, and leakage controls.

Oracle CSV and JSONL fields. Oracle item files use the fields `task_id`, `item_id`, `oracle_label`, `oracle_bucket`, `source_path`, `source_family`, `source_route_stratum`, `reportable`, and, for external repository JSONL files, a representative `line`. Source-route count CSVs use `source_family`, `oracle_bucket`, and `oracle_items`. Oracle labels and oracle totals are loaded only after trajectories, repair orderings, and challenger choices are fixed.

Figure generation. The same supplement assembler writes `main_results_overview.png`, `controller_decision_matrix.png`, and `repair_sensitivity_summary.png` to `2/paper/figures/`. Run the assembler above, then compile from `2/paper/` with `latexmk -pdf -interaction=nonstopmode -outdir=build` and either `evidence_condition_geometry_aaai_v2.tex` or `supplementary_material.tex`.

Reproducibility Protocol

Configuration files. All experiment settings are specified under `2/configs/`. `main_3seed.yaml` is the lightweight reproduction configuration; `full_200seed.yaml` is the paper-grade seeded validation configuration; and `sensitivity.yaml` defines the support/Gini threshold and repair-budget sweeps. These files specify seeds, task sources and routes, repair budgets, τ_s, τ_g, the evaluation-only recall threshold, oracle paths, and output paths.

Runtime versus post-hoc fields. The configs explicitly separate runtime-visible ledger fields from post-hoc oracle-only fields. The controller may use source text, route names, exposure counts, source-route strata, runtime discoveries, costs, and ledger-visible agent notes. It may not use oracle labels, oracle totals, post-hoc recall, hidden missing mass, or undiscovered true-item counts. Oracle fields are loaded only after trajectories, route assignments, repair orderings, and challenger choices have been fixed.

Commands. Set `EVIDENCE_CONFIG=configs/main_3seed.yaml` for a lightweight run or `EVIDENCE_CONFIG=configs/full_200seed.yaml` for the seeded validation reported in the paper. From `2/`, run:

```
P=analysis/research_object_geometry/real_agent_pilot
python $P/scripts/run_blind_policy_task.py
python $P/scripts/run_blind_code_task.py
python $P/external_validation_requests/run_external_requests_validation.py
python $P/external_validation_v2/run_external_validation_v2.py
python $P/credibility_supplement/run_credibility_supplement.py
python paper/scripts/make_paper_figures.py
```

Use `EVIDENCE_CONFIG=configs/sensitivity.yaml` before the supplement assembler for the threshold and budget sweeps.

Outputs and figures. Experiment ledgers are written as JSONL under task-specific `logs/` directories, and aggregate CSV files are written under task-specific `results/` directories and `analysis/research_object_geometry/real_agent_pilot/credibility_supplement/results/`. The supplement assembler also writes a unified CSV/JSON export family: `unified_decision_variants.*`, `unified_repair_variants.*`, `unified_controller_count_table.*`, `unified_localization_risk_trend.*`, `unified_threshold_budget_sweep.*`, `unified_safety_cost_frontier.*`, and `unified_state_metrics.*`. Each unified table carries the task, policy, state type, seed or mixed-seed indicator, budget or mixed-budget indicator, FCR, safe coverage, CONTINUE/ABSTAIN rates, repair gain, and repair cost whenever the metric is defined. `unified_results_manifest.json` records the config path, thresholds, export locations, runtime-visible fields, and post-hoc-only fields. Figure 3 and Figure 4 are regenerated from fixed CSV outputs by `paper/scripts/make_paper_figures.py`, which writes `paper/figures/main_results_overview.pdf`, `paper/figures/evidence_condition_controller.pdf`, and `paper/figures/controller_variant_comparison.pdf`. The same command writes `paper/generated/table_controller_variant_counts.tex`, which is the count table used below.

Controller Certificate Proof Sketch

Proposition 2 in the main paper is a semantic statement about Algorithm 1, not a guarantee that the open world contains no further evidence. The algorithm first computes source-route support and exposure concentration from runtime-visible logs. If the eligibility predicate fails and no repair budget remains, the only possible return is ABSTAIN. If repair is run, any residual evidence found by the fixed runtime-visible repair rule immediately forces CONTINUE. The only branch returning SAFE is reached after eligibility is true and the budgeted repair/audit has produced no residual evidence. Therefore the returned label has the stated bounded meaning: SAFE certifies the declared evidence condition under the fixed budget and runtime-visible repair rule; CONTINUE records productive residual evidence; and ABSTAIN records the absence of a defensible certificate under budget.

Controller Variant Details

Figure 1 compares decision policies on the same fixed bounded-audit states used in the main paper. This is a policy ablation, not a new task suite: oracle labels are used only after decisions are fixed to compute false-SAFE rate, safe-state coverage, decision

Table 1: Controller variant count table. Unsafe denominators are seeded oracle-unsafe repair states; safe denominators are seeded oracle-safe complete states, so these denominators differ from the one-state eligibility boundary in the main text. FCR is SAFE-on-unsafe divided by the unsafe denominator; safe coverage is SAFE-on-safe divided by the safe denominator. Oracle safety labels are used only for post-hoc scoring. Decision variants are evaluated on the residual-potential seeded state set; repair target variants use the same seeds and budgets with their named target rule.

Policy	Unsafe denom.	Safe denom.	SAFE-on-unsafe	SAFE-on-safe	FCR	Safe cov.	CONTINUE	ABSTAIN	Repair gain	Repair cost
Naive stop	400	1200	400	1200	1.000	1.000	0.000	0.000	–	–
Source-only	400	1200	400	1200	1.000	1.000	0.000	0.000	–	–
Verifier-gate	400	1200	0	1200	0.000	1.000	0.000	1.000	–	–
Eligibility-only	400	1200	0	1200	0.000	1.000	0.000	1.000	–	–
Full controller	400	1200	0	1200	0.000	1.000	1.000	0.000	253.0	4275.5
Random repair	400	1200	0	1200	0.000	1.000	0.993	0.007	69.0	3481.1
High-potential repair	400	1200	0	1200	0.000	1.000	1.000	0.000	226.0	3824.0
Ours	400	1200	0	1200	0.000	1.000	1.000	0.000	253.0	4275.5

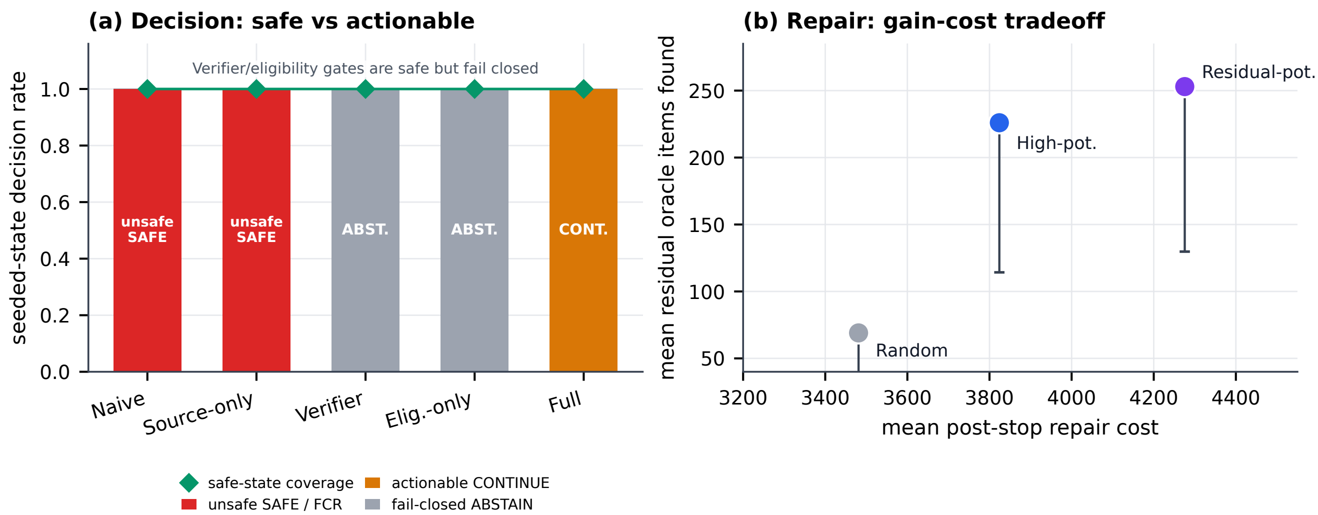

Figure 1: Controller variant comparison. Naive stop acceptance and source-only control accept seeded unsafe states. The full controller and repair-enabled variants avoid SAFE on seeded unsafe repair states while preserving SAFE coverage on seeded complete states. Source-route eligibility alone is evaluated separately on the weak-gap `urllib3` fixed boundary in the main text.

rates, and repair gain/cost. The generic Verifier-gate baseline uses only unresolved-warning and residual-warning bits; it never reads source-route support, Gini, oracle totals, recall, or undiscovered true-item counts. Because those warning bits are explicit runtime signals, Verifier-gate is not allowed to derive warnings from support or Gini. Thus it is a strong fail-closed comparison on seeded residual-warning states, but it is not a replacement for source-route diagnosis on un-warned fixed stops. The main figure reports seeded unsafe-state FCR and seeded safe-state coverage. The observed fixed stop states remain the boundary states from which these seeded validations are derived; they are summarized in Table 2. The main-paper boundary case has a different denominator: it isolates the single eligible-but-residual-positive `urllib3` boundary state. The controller count tables below use seeded unsafe and complete-state denominators. These tables should therefore be read as complementary evidence, not as directly comparable ratios over the same sample.

Computing environment. The checked environment is Windows 11 with Python 3.13.9, NumPy 2.3.5, pandas 2.3.3, and Matplotlib 3.10.6. The code uses only local frozen snapshots for the external repository audits; no network access is required to reproduce the reported tables and figures from the included logs and snapshots.

AAAI Reproducibility Checklist

This paper includes computational experiments. The main text states the bounded scope, assumptions, metrics, thresholds, and controller rule. This supplement records script entry points, seeds, data/log formats, oracle fields, figure-generation commands,

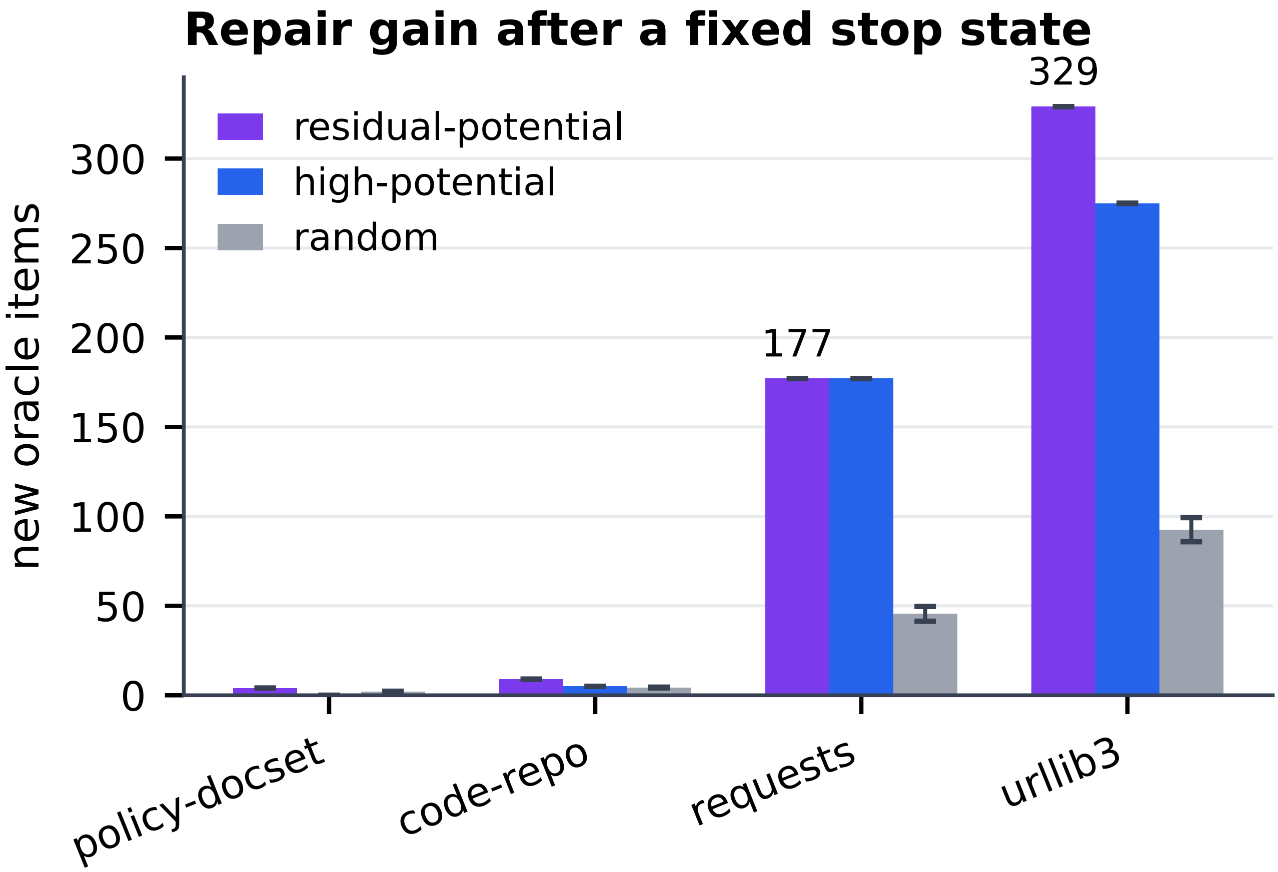

Figure 2: Task-level repair gain for residual-potential, high-potential, and random probes with 95% confidence intervals. This detail complements the main paper's controller-variant comparison and is not a separate task expansion.

software environment, confidence intervals, and threshold/budget sweeps. Randomness is controlled by fixed seed sets before oracle scoring; oracle labels, oracle totals, post-hoc recall, and undiscovered true-item counts are used only for post-hoc scoring. The implementation and generated artifacts are organized as a code/data supplement for anonymous review and can be released on publication.

Table 2: Controller decision accounting. Oracle-safe and oracle-unsafe counts are post-hoc denominators used only for evaluation. Cost is measured repair cost, counted after a fixed repair target set is selected.

Task group	Evaluation set	n	Oracle-safe	Oracle-unsafe	#SAFE	#CONTINUE	#ABSTAIN	False SAFE	FCR	Safe cov.	Abstain	Repair gain	Repair cost
all tasks	observed stop states	9	4	5	4	2	3	0	0.000	1.000	0.333	0.000	0.0
external repos	observed external states	5	2	3	2	2	1	0	0.000	1.000	0.200	0.000	0.0
requests	seeded repairs	1200	0	1200	0	1199	1	0	0.000	–	0.001	122.992	2854.6
urllib3	seeded repairs	1000	0	1000	0	998	2	0	0.000	–	0.002	204.987	4831.8
external repos	seeded repairs	2200	0	2200	0	2197	3	0	0.000	–	0.001	160.262	3753.3
requests	seeded safe states	3000	3000	0	3000	0	0	0	0.000	1.000	0.000	0.000	3252.5
urllib3	seeded safe states	3000	3000	0	3000	0	0	0	0.000	1.000	0.000	0.000	3933.7
external repos	seeded safe states	6000	6000	0	6000	0	0	0	0.000	1.000	0.000	0.000	3593.1

Table 3: Per-task decision breakdown on observed stop states. FCR is SAFE-on-unsafe divided by oracle-unsafe stop states for that task; safe coverage is SAFE-on-safe divided by oracle-safe stop states for that task. Oracle labels are post-hoc only.

Task	Policy	Unsafe	Safe	SAFE-on-unsafe	FCR	Safe cov.	CONTINUE	ABSTAIN
policy-docset	Naive stop	1	1	1	1.000	1.000	0.000	0.000
policy-docset	Source-only	1	1	1	1.000	1.000	0.000	0.000
policy-docset	Verifier-gate	1	1	1	1.000	1.000	0.000	0.000
policy-docset	Eligibility-only	1	1	0	0.000	1.000	0.000	0.500
policy-docset	Full controller	1	1	0	0.000	1.000	0.000	0.500
code-repo	Naive stop	1	1	1	1.000	1.000	0.000	0.000
code-repo	Source-only	1	1	1	1.000	1.000	0.000	0.000
code-repo	Verifier-gate	1	1	1	1.000	1.000	0.000	0.000
code-repo	Eligibility-only	1	1	0	0.000	1.000	0.000	0.500
code-repo	Full controller	1	1	0	0.000	1.000	0.000	0.500
requests	Naive stop	1	1	1	1.000	1.000	0.000	0.000
requests	Source-only	1	1	1	1.000	1.000	0.000	0.000
requests	Verifier-gate	1	1	1	1.000	1.000	0.000	0.000
requests	Eligibility-only	1	1	0	0.000	1.000	0.000	0.500
requests	Full controller	1	1	0	0.000	1.000	0.000	0.500
urllib3	Naive stop	2	1	2	1.000	1.000	0.000	0.000
urllib3	Source-only	2	1	2	1.000	1.000	0.000	0.000
urllib3	Verifier-gate	2	1	2	1.000	1.000	0.000	0.000
urllib3	Eligibility-only	2	1	1	0.500	1.000	0.000	0.333
urllib3	Full controller	2	1	0	0.000	1.000	0.667	0.000

Table 4: Lightweight external baselines and sanity checks. These are not new task suites: source-only is a direct stopping baseline, random/high-potential are repair-target baselines under the same budget, and the scalar singleton/Chao proxy is a negative control that ignores source-route geometry.

Task	Baseline	Measurement	Value	Note
requests	Source-only stop	FCR if accepted	1.000	accepts localized route evidence
requests	random expansion	mean repair gain	45.5	same budget, post-stop repair
requests	high-potential expansion	mean repair gain	177.0	same budget, post-stop repair
requests	residual-potential expansion	mean repair gain	177.0	same budget, post-stop repair
urllib3	Source-only stop	FCR if accepted	1.000	accepts localized route evidence
urllib3	random expansion	mean repair gain	92.5	same budget, post-stop repair
urllib3	high-potential expansion	mean repair gain	275.0	same budget, post-stop repair
urllib3	residual-potential expansion	mean repair gain	329.0	same budget, post-stop repair
requests	Singleton/Chao scalar stop	would stop?	no	Chao1=496.0
urllib3	Singleton/Chao scalar stop	would stop?	no	Chao1=9180.0

Table 5: Negative control: scalar singleton/Chao proxy on homogeneous runs. The proxy observes item-count sparsity but does not encode source-route condition, so it is not a competitive controller baseline for certificate mismatch.

Task	Observed	Oracle total	Recall	f_1	f_2	f_1/obs	Chao1	Scalar stop	False if stop
policy_docset_v1	17	24	0.708	17	0	1.000	153.0	No	No
code_repo_v1	6	20	0.300	6	0	1.000	21.0	No	No
requests	31	298	0.104	31	0	1.000	496.0	No	No
urllib3	135	699	0.193	135	0	1.000	9180.0	No	No

Table 6: Sensitivity sweep numeric summary. Ranges aggregate threshold settings or repair-budget settings within each external task.

Task	Sweep	SAFE rate	CONTINUE rate	ABSTAIN rate	Max FCR	Mean repair cost
requests	threshold	0.000-0.000	0.995-1.000	0.000-0.005	0.000	2854.6
urllib3	threshold	0.000-0.000	0.990-1.000	0.000-0.010	0.000	4831.8
requests	budget	0.000-0.000	0.635-1.000	0.000-0.365	0.000	3179.0
urllib3	budget	0.000-0.000	0.640-1.000	0.000-0.360	0.000	4323.5

Table 7: External oracle construction summary. Pattern-defined external oracles are line-level source-route evidence occurrences from frozen local snapshots.

Repo	Version	Items	Routes	Nonempty strata	Granularity
requests	2.32.5	298	4	18	line-level source-route
urllib3	2.5.0	699	5	24	line-level source-route

Table 8: Small logged workflow pilot for workflow-shape/interface validation only. The pilot fixes the prompt and records independent contexts, action/evidence events, localized stop proposals, and the controller-facing decision record; it is not used as main effectiveness evidence.

Task	Model	Agents	Independent contexts	Action events	Evidence events	Stop proposals	Controller decision
T_doc_dynamic_workflow_smoke	logged LLM-agent workflow	3	3	39	35	3	not scored; workflow-shape validation

Table 9: Seeded safe-state validation. These are oracle-safe broad-complete states with seeded order perturbations; the controller should remain SAFE rather than over-reacting to extra repair opportunity.

Task	State	Order/budget sweep	Runs	#SAFE	#CONTINUE	#ABSTAIN	FCR	Safe cov.	Repair gain	Mean cost	Cost range
requests	route_partitioned	5 orders; budgets 1-8	3000	3000	0	0	0.000	1.000	0.0	3252.5	157-7647
urllib3	extended_audit	5 orders; budgets 1-8	3000	3000	0	0	0.000	1.000	0.0	3933.7	275-9072

Table 10: Route-match consistency sanity check for external oracle positives. A small sample of line-level positives was checked for route consistency; ambiguous lines are retained as such rather than treated as full validation.

Repo	Route	Sample	Positive	Precision	Ambiguous	Notes
requests	compat_route	8	8	1.000	0	line-level oracle-positive sample
requests	exception_route	8	8	1.000	0	line-level oracle-positive sample
requests	timeout_route	8	8	1.000	0	line-level oracle-positive sample
requests	tls_route	8	8	1.000	0	line-level oracle-positive sample
urllib3	cleanup_route	8	7	0.875	1	line-level oracle-positive sample
urllib3	exception_route	8	8	1.000	0	line-level oracle-positive sample
urllib3	retry_route	8	8	1.000	0	line-level oracle-positive sample
urllib3	timeout_route	8	8	1.000	0	line-level oracle-positive sample
urllib3	tls_route	8	8	1.000	0	line-level oracle-positive sample

Table 11: Leakage control for controller and repair decisions. Runtime decisions use only visible evidence. Oracle labels, oracle totals, post-hoc recall, and undiscovered true item counts are used only after trajectories and challenger choices are fixed.

Component	Runtime allowed	Oracle forbidden
Exposure	visits, scans, actions	missing mass
Support/Gini	exposure counts	post-hoc recall
Potential	text, routes, matches	hidden true items
Decision	condition and repair	target distribution

Table 12: External oracle appendix detail. Each route uses a frozen regex pattern; items are deduplicated by source, route, and line. To avoid unreadable regex truncation, the appendix reports a short pattern category and one representative line-level item per route; full regex rules and items are in the generated CSVs.

Repo	Route	Items	Pattern category	Dedup	Representative item
`requests`	`tls_route`	102	TLS/certificate/verification terms	source+route+line	adapters:tls_route:103: # cert path
`requests`	`timeout_route`	31	timeout/connect/read-timeout terms	source+route+line	adapters:timeout_route:120: self, request, stream=False, timeout=None, verify=True, cert=None, proxies=None
`requests`	`exception_route`	121	exception/raise/error terms	source+route+line	adapters:exception_route:136: raise NotImplementedError
`requests`	`compat_route`	44	compatibility/deprecation terms	source+route+line	adapters:compat_route:32: from .compat import basestring, urlparse
`urllib3`	`timeout_route`	135	timeout/connect/read-timeout terms	source+route+line	connection:timeout_route:137: timeout: _TYPE_TIMEOUT = _DEFAULT_TIMEOUT,
`urllib3`	`retry_route`	168	retry/backoff/status terms	source+route+line	connectionpool:retry_route:1011: retries,
`urllib3`	`tls_route`	111	TLS/certificate/verification terms	source+route+line	connection:tls_route:1005: cert,
`urllib3`	`exception_route`	170	exception/raise/error terms	source+route+line	connection:exception_route:1015: except BaseException:
`urllib3`	`cleanup_route`	115	close/release/finally terms	source+route+line	connection:cleanup_route:1016: ssl_sock.close()